HHCKLA Buddhist Leung Chik Wai College

6A error analysis (2026)

Student No.: 08 Name: _____ Marks: / 54

Class: 6A Date: _____ Marked date: _____

What have you learned in this chapter? _____

What can you do to improve your work next time? _____

1. The sum of the number of blue pens and the number of red pens in a box is 795. If 123 blue pens are taken out from the box, then the number of blue pens in the box is 25% less than the number of red pens in the box. Find the number of red pens in the box. (4 marks) (DSE-2026 PI Q05)

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2. The stem-and-leaf diagram below shows the distribution of the numbers of driving hours of a group of taxi drivers in a week.

Stem (tens)	Leaf (units)
2	5 6 9
3	2 3 3 6 6 7 8
4	0 0 2 2 5 5 7 7 7
5	w

The ratio of the range to the inter-quartile range of the distribution is 7 : 3.

(a) Find w.

(b) (i) Write down the mode of the distribution.

(ii) If two taxi drivers are randomly selected from the group, find the probability that the mean of their numbers of driving hours in the week exceeds the mode of the distribution.

(6 marks) (DSE-2026 PI Q10)

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3. The straight line L passes through the point $(-10, 11)$. The slope of L is -2.

(a) Find the equation of L.

(b) The coordinates of the points H and K are $(2, 5)$ and $(-4, -7)$ respectively. Let P be a point lying on L such that P is equidistant from H and K.

(i) Find the coordinates of P.

(ii) Let Q be a point such that $HPKQ$ is a quadrilateral, where $HP = HQ = KQ$. Find the area of the quadrilateral $HPKQ$.

(7 marks) (DSE-2026 PI Q12)

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4. It is given that P, Q and R are three points in a rectangular coordinate plane. The coordinates of P and Q are $(7,8)$ and $(-13,20)$ respectively. The equation of the perpendicular bisector of PR is $3x+5y-27=0$. Let C be the circle which passes through P, Q and R.

(a) Find the equation of C.

(b) The tangent to C at P and the tangent to C at R intersect at the point H. Is $\triangle HPR$ a right-angled triangle? Explain your answer.

(6 marks) (DSE-2026 PI Q16)

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5. Let $G(n)$ be the nth term of a geometric sequence. It is given that $G(2) = 12$ and $G(5) = 768$.

(a) Find $G(1)$.

(b) Define $T(n) = \log_2 G(n)$ for all positive integers n. Find

(i) $T(n+1) - T(n)$,

(ii) the greatest value of k such that $T(1) + T(2) + T(3) + \cdots + T(k) < 10^6$.

(7 marks) (DSE-2026 PI Q18)

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6. Let $h(x) = -x^2 + 30kx + 4x - 225k^2 - 40k + 3$, where k is a positive constant. On the same rectangular coordinate system, denote the vertex of the graph of $y = h(x)$ and the vertex of the graph of $y = 14 - h(4-x)$ by A and B respectively.

(a) Using the method of completing the square, express, in terms of k, the coordinates of A.

(b) Write down, in terms of k, the coordinates of B.

(c) The coordinates of the point C are $(2 + 48k,\ 7 - 36k)$. Let I and J be the in-centre and the circumcentre of $\triangle ABC$ respectively.

(i) Let M be the mid-point of AB. Describe the geometric relationship between CM and AB. Explain your answer.

(ii) Express, in terms of k, the distance between I and J.

(iii) The coordinates of the point D are $(6 - 22k,\ 6 - 15k)$. Is it possible that $BDIJ$ is a parallelogram? Explain your answer.

(12 marks) (DSE-2026 PI Q19)

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7. The perimeter and the area of a sector are k cm and $3k$ cm^2 respectively. If the radius of the sector is 14 cm, then the arc length of the sector is

A. 19 cm.

B. 21 cm.

C. 28 cm.

D. 49 cm.

(DSE-2026 PII Q16)

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8. It is given that $WXYZ$ is a quadrilateral, where $\angle WXY = \angle YWZ$ and $\angle WYX = \angle WZY$. If $WX = 9$ cm, $WY = 40$ cm and $XY = 41$ cm, find the perimeter of the quadrilateral $WXYZ$.

A. 100 cm

B. 180 cm

C. 410 cm

D. 490 cm

(DSE-2026 PII Q21)

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9. It is given that $ABCDEF$ is a regular 6-sided polygon. If AC and BF intersect at the point G, which of the following are true?

I. $BG//CE$

II. $\triangle ABG \sim \triangle BDC$

III. $\triangle AGF \cong \triangle BGC$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

(DSE-2026 PII Q23)

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10. A bag contains five cards numbered 2, 2, 3, 4 and 6 respectively. Two cards are randomly drawn from the bag at the same time. Find the probability that the sum of the numbers drawn is greater than 7.

A. $\frac{1}{2}$

B. $\frac{1}{5}$

C. $\frac{2}{5}$

D. $\frac{3}{10}$

(DSE-2026 PII Q28)

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11. Consider the following system of inequalities:

$$\begin{cases} y \leq 7 \\ 3x - 4y \leq 2 \\ 6x + 13y \geq 25 \end{cases}$$

Let R be the region which represents the solution of the above system of inequalities. If (x, y) is a point lying in R, then the least value of $5x + 4y + 28$ is

A. 1.

B. 19.

C. 42.

D. 106.

(DSE-2026 PII Q37)

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12. There are 5 full-time employees and 3 part-time employees in a company. From the company, 7 employees are selected to form a queue. If no part-time employees are next to each other, how many different queues can be formed?

A. 5040

B. 14400

C. 18000

D. 32400

(DSE-2026 PII Q43)

I can finish _____ questions

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